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Exercise Sheet 3

This is not the official exercise sheet from the course.
More difficult exercises are denoted by ★ to ★★★

If you can solve it, it is an exercise;
otherwise it's a research problem.

Richard Bellman

Exercise 1.
For each of the following sets A and B, decide which of $A \in B, A \subseteq B, A \ni B, A \supseteq B$ is true.

1. $A = \{\{1,2\},3\}$, $B = \{1,2\}$
2. $A = \mathbb{N}$, $B = \mathbb{R}$
3. $A = \{\{\emptyset\},\emptyset\}$, $B = \{\emptyset,\{\emptyset\}\}$
4. $A = \{x \in \mathbb{R} : x^2 - 4 = 0\}$, $B = \{x \in \mathbb{Z} : |x| \leq 2\}$
5. $A = \{\emptyset,\{\emptyset\}\}$, $B = \emptyset$
6. $A = \{\emptyset,\{\emptyset\},\{\{\emptyset\}\},\{\emptyset,\{\emptyset\}\}\}, B = \{\emptyset,\{\emptyset\}\}$
7. $A = \emptyset$, $B = \mathbb{N}$
8. $A = \{\emptyset\}$, $B = \{\emptyset,\{\emptyset\}\}$
9. $A = \{\emptyset,\{\emptyset\}\}$, $B = \{\{\emptyset\}\}$
10. $A = \mathcal{P}(\emptyset)$, $B = \mathcal{P}(\mathcal{P}(\emptyset))$
11. $A = \{\emptyset\}$, $B = \mathcal{P}(\{\emptyset\})$
12. $A = \{\{\emptyset,\{\emptyset\}\}\}$, $B = \{\emptyset,\{\emptyset\},\{\emptyset,\{\emptyset\}\}\}$
13. $A = \{\emptyset \setminus \{\emptyset\}\}$, $B = \emptyset$
14. $A = \mathcal{P}(\emptyset) \setminus \emptyset$, $B = \{\emptyset\} \setminus \{\emptyset\}$

Solution 1.

1. $A \ni B$ is true (since $B = \{1, 2\} \in A$).
2. $A \subseteq B$ is true (since $\mathbb{N} \subseteq \mathbb{R}$).
3. $A \subseteq B$ and $A \supseteq B$ are true (since $A = B$).
4. $A \subseteq B$ is true (since $A = \{-2, 2\}$ and $B = \{-2, -1, 0, 1, 2\}$).
5. $A \ni B$ and $A \supseteq B$ are true (since $B = \emptyset \in A$ and $\emptyset \subseteq A$).
6. $A \ni B$ and $A \supseteq B$ are true (since $B = \{\emptyset, \{\emptyset\}\} \in A$ and $B \subseteq A$).
7. $A \subseteq B$ is true (since $\emptyset \subseteq \mathbb{N}$).
8. $A \in B$ and $A \subseteq B$ are true (since $\{\emptyset\} \in B$ and $\emptyset \in B$).
9. $A \supseteq B$ is true (since $B = \{\{\emptyset\}\}$ and $B \subseteq A$).
10. $A \in B$ and $A \subseteq B$ are true (since $\mathcal{P}(\emptyset) = \{\emptyset\}$ and $\mathcal{P}(\mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\}$).
11. $A \in B$ and $A \subseteq B$ are true (since $A = \{\emptyset\}$ and $B = \mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$).
12. $A \subseteq B$ is true.
13. $A \ni B$ and $A \supseteq B$ are true (since $\emptyset \setminus \{\emptyset\} = \emptyset$, making $A = \{\emptyset\}$ and $B = \emptyset$).
14. $A \ni B$ and $A \supseteq B$ are true (since $\mathcal{P}(\emptyset) \setminus \emptyset = \{\emptyset\}$ and $\{\emptyset\} \setminus \{\emptyset\} = \emptyset$).

Exercise 2.
Let $|A| = n$ and $|B| = m$. Give the cardinalities of the following sets in terms of n, m.

1. $|\mathcal{P}(\mathcal{P}(\mathcal{P}(A)))|$
2. $|\mathcal{P}(A \times B)|$
3. $|\mathcal{P}(A) \times \{\emptyset\}|$
4. $|\mathcal{P}(\mathcal{P}(\mathcal{P}(A \times \emptyset)))|$
5. $|\mathcal{P}(\mathcal{P}(A)) \times \mathcal{P}(\mathcal{P}(B))|$
6. $|(\mathcal{P}(A) \times B) \setminus \emptyset|$

Solution 2.

1. $2^{2^{2^n}}$
2. $2^{n \cdot m}$
3. 2^n
4. 4
5. $2^{2^n + 2^m}$
6. $2^n \cdot m$

Exercise 3.
Let A, B be arbitrary sets. Prove or disprove each of the following statements.

1. $\mathcal{P}(A \times B) = \mathcal{P}(A) \times \mathcal{P}(B)$
2. $\mathcal{P}(A \setminus B) = \mathcal{P}(A) \setminus \mathcal{P}(B)$
3. If $\mathcal{P}(A) \subseteq \mathcal{P}(B)$, then $A \subseteq B$

Solution 3.

1. *Proof.* This is false. Consider the following counterexample. Let $A = \{1\}$ and $B = \{2\}$. Then $A \times B = \{(1, 2)\}$, so $\mathcal{P}(A \times B) = \{\emptyset, \{(1, 2)\}\}$, which has cardinality 2. On the other hand, $\mathcal{P}(A) = \{\emptyset, \{1\}\}$ and $\mathcal{P}(B) = \{\emptyset, \{2\}\}$, so $\mathcal{P}(A) \times \mathcal{P}(B)$ has cardinality $2 \cdot 2 = 4$. Since their cardinalities differ, $\mathcal{P}(A \times B) \neq \mathcal{P}(A) \times \mathcal{P}(B)$. □

2. *Proof.* This is false. Consider the following counterexample. Let $A = \{1\}$ and $B = \{1\}$. Then $A \setminus B = \emptyset$, so $\mathcal{P}(A \setminus B) = \mathcal{P}(\emptyset) = \{\emptyset\}$. However, $\mathcal{P}(A) = \{\emptyset, \{1\}\}$ and $\mathcal{P}(B) = \{\emptyset, \{1\}\}$, giving $\mathcal{P}(A) \setminus \mathcal{P}(B) = \emptyset$. Since $\{\emptyset\} \neq \emptyset$, the statement is false. □

3. *Proof.* To prove that if $\mathcal{P}(A) \subseteq \mathcal{P}(B)$, then $A \subseteq B$, assume $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. Let x be an arbitrary element such that $x \in A$. Then the singleton set $\{x\}$ is a subset of A, meaning $\{x\} \in \mathcal{P}(A)$. Since $\mathcal{P}(A) \subseteq \mathcal{P}(B)$, it follows that $\{x\} \in \mathcal{P}(B)$. By the definition of the power set, $\{x\} \subseteq B$, which implies $x \in B$. Therefore, every element of A is an element of B, so $A \subseteq B$. □

Exercise 4.

Let A, B be finite sets. Prove or disprove each of the following statements.

1. $|\mathcal{P}(A \cup B)| = |\mathcal{P}(A)| + |\mathcal{P}(B)| - |\mathcal{P}(A \cap B)|$
2. $|\mathcal{P}(A \times B)| = 2^{|A| \cdot |B|}$
3. If $|\mathcal{P}(A)| = |\mathcal{P}(B)|$, then $A = B$

Solution 4.

1. *Proof.* This is false. Consider the following counterexample. Let $A = \{1\}$ and $B = \{2\}$. Then $A \cup B = \{1, 2\}$, so $|\mathcal{P}(A \cup B)| = 2^2 = 4$. However, $|\mathcal{P}(A)| + |\mathcal{P}(B)| - |\mathcal{P}(A \cap B)| = 2^1 + 2^1 - 2^0 = 2 + 2 - 1 = 3$. Since $4 \neq 3$, the equality does not hold. □

2. *Proof.* To prove $|\mathcal{P}(A \times B)| = 2^{|A| \cdot |B|}$, let $|A| = n$ and $|B| = m$. The cardinality of the Cartesian product of two finite sets is the product of their cardinalities, so $|A \times B| = |A| \cdot |B| = n \cdot m$. The cardinality of the power set of any finite set X is given by $2^{|X|}$. Substituting $A \times B$ for X yields $|\mathcal{P}(A \times B)| = 2^{|A \times B|} = 2^{n \cdot m} = 2^{|A| \cdot |B|}$. This concludes the proof. □

3. *Proof.* This is false. Consider the following counterexample. Let $A = \{1\}$ and $B = \{2\}$. Then $|\mathcal{P}(A)| = 2^1 = 2$ and $|\mathcal{P}(B)| = 2^1 = 2$, so $|\mathcal{P}(A)| = |\mathcal{P}(B)|$. However, $A \neq B$ because $1 \in A$ but $1 \notin B$. □

★ **Exercise 5.**

Let Let $A_n = \{n\}$ for each $n \in \mathbb{N}$ be a sequence of sets and $\mathbb{P}$ the set of all prime numbers. Compute

1. $\bigcup_{n=0}^{\infty} A_n$
2. $\bigcap_{n=0}^{\infty} A_n$
3. $\bigcap_{n=0}^{\infty} A_{3n}$
4. $\left(\bigcup_{n\in\mathbb{P}} A_n\right) \setminus \left(\bigcup_{n\in\mathbb{N}\setminus\mathbb{P}} A_n\right)$
5. $\bigcap_{n=0}^{\infty} (A_n \cup A_{n+1})$
6. $\bigcup_{n=1}^{\infty} (A_n \setminus A_{n-1})$

★ **Solution 5.**

1. $\mathbb{N}$
2. $\emptyset$
3. $\emptyset$
4. $\mathbb{P}$
5. $\emptyset$
6. $\mathbb{N} \setminus \{0\}$